Tech Stack

Smart Ambulance Green Corridor System

1. Core Logic

Block	Component	Purpose
Logic family	74HC series, 5 V	All digital logic
System clock	NE555 astable, 1 Hz (pot adjustable)	Drives synchronizers, counter, FSM
Priority encoder	74HC148	Selects the highest-priority request direction
FSM state storage	74HC74 flip-flops	Holds current state
Next-state logic	74HC00 / 08 / 32 / 04 gates	Computes next state
Timing counter	74HC161 (4-bit)	Measures green, yellow, all-red and timeout durations
Counter decode	Gate network	Detects terminal counts for 1, 2, 8, 10 s

2. Switch Inputs (Ambulance Arrival)

Block	Component	Purpose
Request switches	4 push buttons (REQ_N, REQ_E, REQ_S, REQ_W)	Pressed to simulate an ambulance arriving on that approach
Clear switch	1 push button (CLEAR)	Simulates the ambulance having crossed the junction
Reset switch	1 push button or toggle	Returns FSM to NS_G and clears latches
Debounce	10 k + 1 uF RC filter, 74HC14 Schmitt inverter	Removes contact bounce (about 10 ms time constant)
Request latch	74HC00 NAND SR latch (x4)	Holds request after the button is released
Clear routing	74HC139 decoder	Resets only the latch of the direction being served
Synchronizer	74HC74 D flip-flops	Aligns requests to the system clock

3. Output and Power

Block	Component	Purpose
Lamp decoder	74HC138 / 139	Converts state to lamp select lines
Indicators	12 LEDs (R/Y/G x 4) + 330 ohm resistors	Traffic signals
Driver (optional)	ULN2803	For bulbs or higher current loads
Regulator	7805 (5 V)	Supply regulation
Source	9 V adapter	Input power
Decoupling	0.1 uF ceramic per IC	Noise suppression

4. Tools and Build Platform

- **Simulation:** Logisim or Proteus
- **Prototyping:** breadboard, then perfboard

• **Test equipment:** multimeter, logic probe, oscilloscope (if available)

• **Model:** small road-layout board with a switch panel for the four approaches

5. Why This Stack

• 74HC runs at 5 V with low power and tolerates breadboard wiring well

• Push buttons with latches are simple, reliable, and easy to demonstrate

• A 555 timer gives the slow clock with no code

• Every block maps directly to the project's required topics: counters, priority encoder, FSM